

Manufacturing Sourcing Guide for "Tau" — A Low-Volume Magnetic Balance Board Game

TL;DR

- **At 10–500 units, keep every component in the "on-demand / no-tooling" regime:** MJF 3D-print the tripod pieces, laser-cut 430 stainless discs finished by a separate UV flatbed printer, buy stock neodymium magnets in bulk, and order a rigid two-piece (telescoping/hatbox) box with a die-cut cardboard insert from a game-focused print-on-demand or low-MOQ packaging house. Do **not** tool up (injection molds, stamped/litho metal boards, custom pulp molds) until you are confident of runs in the low thousands.
- **The cheap-travel-chess trick (litho-printed tin-plated steel boards + injection-molded magnetic plastic pieces) is a pure high-volume play** — offset lithography on tinplate has a minimum of about 1,500 printed sheets, and full custom tin tooling starts at 5,000 units for existing sizes and 10,000 for custom sizes. It is not accessible or economic at your volumes; the low-volume equivalent is laser-cut steel + UV flatbed printing.
- **Taiwan is a genuine strategic asset:** it has dense clusters of small-batch injection molders (MOQ ~1,000), metal stampers, and mold makers used to export/crowdfunding clients, plus tinplate litho printers offering small-quantity gang-run printing. Use your Taipei/Taichung family contacts for QC, assembly consolidation, and to bridge to tooled production later.

Key Findings

1. Crossover volumes for this specific product. For the nylon tripod pieces, MJF 3D printing stays cheaper than injection molding until roughly 1,000 units for a part this size; below that, tooling never pays back. For the metal board, laser-cutting + UV printing beats stamped/litho tinplate until you are ordering many thousands of boards. For the box and insert, print-on-demand / low-MOQ cardboard is competitive up to a few hundred units, after which offset-printed rigid boxes with real dielines from an overseas game manufacturer win. Moulded pulp is essentially a future-only option because of custom tooling cost.

2. The single biggest decision is the metal board. No online service does "laser-cut ferrous disc AND print the colored rings" in one step. You will run a two-vendor chain: (a) a laser cutter produces the 430 stainless disc; (b) a UV flatbed printer lays down the concentric colored rings directly on the steel. Both steps are available at low/no minimum in Australia and Taiwan.

3. Magnets and rules cards are trivially solved and cheap at any volume. Neodymium discs are a stock commodity; rules/QR cards are standard print-on-demand.

Details by Category

A. How cheap travel magnetic chess/checkers sets achieve low unit cost — and why it doesn't transfer to low volume

The classic \$10–20 folding travel set is a textbook high-volume Chinese contract-manufacturing product. The board is typically thin tin-plated or painted steel (or a steel foil laminated inside a plastic/vinyl folding case), decorated by **offset lithography printed on the flat metal sheet before it is cut and formed** — the same chromolithography-on-tinplate process used for tin toys since the 1880s. The pieces are injection-molded ABS or HIPS plastic with a small magnet (or magnetized ferrite disc) embedded or glued into the base; many budget sets instead use a plain ferrite/rubber magnetic pad on the piece base with a plain steel board.

Why it's cheap: essentially all the cost sits in tooling and setup that is amortized over tens of thousands of units. The critical constraint for you is the printing step. Per tinplate printer Tinspkg, "The printing minimum quantity is 1500 pieces tinplate sheet and sheet size is generally 1085×960mm." And PakFactory states for custom tins: "Currently, our MOQ [is] 5000 units for existing sizes and 10,000 for custom sizes. Molds for tin boxes are extremely time consuming and expensive to create." Some Chinese/Taiwanese tin printers (e.g., Tinspkg; Taiyeng in Taiwan) offer a workaround — **gang-run printing**, where many customers' designs share one tinplate sheet to split the setup cost — which can pull effective minimums down toward the low thousands, but still far above 500.

Conclusion: The litho-tinplate technique is not viable at 10–500 units. The correct low-volume analog is **laser/water-jet-cut steel + digital UV flatbed printing**, which has no tooling and no meaningful minimum — you pay per part and per print. You lose economy of scale but gain zero setup cost and full design freedom, which is exactly right at your volumes.

B. Magnetic metal board (~267 mm / 430 stainless disc + printed rings)

Material choice. 430 stainless is the correct call: it is one of the few stainless grades that is ferritic (magnetic), unlike the far more common 304/316 (which are largely non-magnetic and are what most laser shops stock by default). Confirm any quote is specifically 430 (or another ferritic/mild steel), not 304/316. Mild (carbon) steel is also magnetic and cheaper but needs a coating/print layer to prevent rust; 430 stainless resists corrosion without plating, which matters for a bare gameplay surface.

Step 1 — Cutting the disc:

- **Ponoko** (online, ships to Australia) explicitly offers **430 stainless steel laser cutting** with **no minimum order** — its metal page lists 430 stainless "From: \$1.22" in 0.91 mm, 1.22 mm and 1.52 mm thicknesses, with shipping "from just \$50." You can order a single disc; this is the most turnkey option for the exact grade you want, and the 0.91–1.52 mm range brackets the gauge you'll want for magnet grip.
- **SendCutSend** (US) — instant online quoting, "order one part or ten thousand," ± 0.005 " tolerance, discounts up to ~80% in bulk, ships internationally. Stocks primarily 304/316 stainless; confirm 430 availability or accept a magnetic mild-steel alternative.
- **Australian local shops** for domestic turnaround and no import friction: Australian Laser Cut (auslaser.com.au, ships Australia-wide, cuts 304/316), Hygrade Laser Profiling (Sydney), SMS Laser Cutting (Sydney + Melbourne), Lightning Laser (Melbourne, cuts stainless 0.1–16 mm), KNS Metals (Melbourne, stainless up to 12 mm). Most quote per-file and accept one-offs and small runs; you must specifically request 430/ferritic stock.
- **Xometry / Fictiv / OSH Cut** — all do instant-quote sheet metal but, like SendCutSend, default to 304/316; verify grade.
- A ~267 mm disc in ~0.8–1.5 mm steel is small and simple, so per-unit cutting cost is low and drops sharply with quantity (nesting many discs per sheet).

Step 2 — Printing the rings (a separate vendor):

- **UV flatbed printing** deposits UV-cured ink directly onto metal with no screens/tooling, making it ideal for short runs. Providers describe it as explicitly cost-effective for prototypes and low volume (EVS Metal, Maas Brothers in the US; many Australian sign/print shops run Roland or Mimaki flatbeds). Search local "UV flatbed printing on metal" sign shops in Melbourne/Sydney — this is a common signage capability.
- Alternative: **screen printing** on the discs becomes cheaper than UV only at higher volumes (hundreds+ of the same design) because it needs a screen per color; for 10–500 with multiple ring colors, UV flatbed is superior.
- **Adhesive-print alternative (cheapest to prototype)**: print the graphic on a circular vinyl/paper laminate label and apply to the steel. This is effectively what many low-cost boards do; it lets you skip UV printing entirely at the cost of a less premium finish and a potential wear/peel point.

C. Plastic tripod pieces (11–15 g, chess-piece size) via MJF

MJF (HP Multi Jet Fusion) PA12 nylon is the right process: strong, isotropic, no support marks, no tooling, and excellent at small-batch. Because MJF fills a whole powder bed per build, cost drops as you nest more parts and then flattens once the bed is full — for a part this small that flattening happens in the low hundreds of units.

Concrete pricing (from real dated quotes + industry ranges; treat per-tier estimates as directional):

- A single small MJF part from premium Western services (Xometry, Protolabs, Materialise) effectively runs **~\$40–\$130** because of fixed per-order/handling economics; the same part

from **Craftcloud** (an aggregator) or **JLC3DP** can be well under \$40 for one.

- Real July 2024 quotes (All3DP, "MJF 3D Printing Service Costs: 5 Projects Compared," Jul 11 2024) for a single small natural-grey nylon MJF part (spiral-sphere container): **Craftcloud \$39.03, Forecast 3D \$50, Materialise OnSite \$85.13, Xometry \$129.12.**
- Same comparison, **300 units** of a small part (cable organizer, PA12, vapor-smoothed and black-dyed): **Craftcloud \$982.73 (~\$3.28/ea), Materialise OnSite \$1,919.70 (~\$6.40/ea), Xometry \$2,490 (~\$8.30/ea), Forecast 3D \$2,913 (~\$9.71/ea), Protolabs \$4,209 (~\$14.03/ea)** — note these include premium finishing, so plain natural-grey parts would be cheaper.
- Rough per-tier estimate for your ~11–15 g part, natural grey, no premium finish: **1 unit ~ \$15–\$40; 10 ~ \$8–\$20 each; 50 ~ \$5–\$12; 100 ~ \$4–\$9; 500 ~ \$3–\$7.** With two pieces per game, budget roughly **\$6–\$18/game for the pair at small runs, falling toward ~\$6–\$8/pair at 500 games.**
- Material cost is trivial (PA12 powder ~\$50–\$80/kg, [Unionfab](#) ↗ so ~\$1 of powder in the part); price is dominated by machine time, handling, and margin.

Named services with instant online quoting: Xometry, Protolabs Network (formerly Hubs), Craftcloud (aggregates 150+ shops [Craftcloud®](#) ↗

— usually cheapest), JLC3DP / JLCPCB (from ~\$1/part base, [JLC3DP](#) ↗

ships from China, no minimum), PCBWay, Shapeways, plus Australian MJF bureaus.

Recommendation: Get instant quotes from Craftcloud, JLC3DP, and one Australian bureau simultaneously; Craftcloud/JLC3DP for price, a local bureau for speed and iteration. Most advertise no MOQ (the sometimes-cited "\$29 Xometry minimum" is not confirmed in current sources — Xometry officially states no minimum order).

Note on the magnet-in-foot design: MJF can print small pockets/cavities in the tripod feet to press-fit or glue the neodymium magnets. Design these cavities with ≥ 0.4 mm clearance for MJF tolerances. [Xometry Pro](#) ↗

D. Neodymium magnets (small discs, thousands)

A pure commodity. For gameplay-grade grip you likely want small N42/N45 discs around 5–10 mm dia × 1–3 mm. Bulk suppliers:

- **K&J Magnetics** (US) — huge inventory, volume discounts, engineering support and pull-force calculators.
- **totalElement, Applied Magnets, CMS Magnetics, Magnet4Sale** (US) — all bulk/wholesale.
- **HSMAG / Stanford Magnets** (China) — factory-direct, custom sizes, cheapest at thousands.
- **Australian:** Frenergy, AMF Magnets, Magnet Shop (AU) for domestic supply without import hassle.

- Standard Ni-Cu-Ni triple coating; expect a few cents each at thousands. Choose grade/thickness by testing grip on your actual board thickness — magnet pull through steel depends on board gauge. Budget well under \$0.50/game in magnets.

E. Box — octagonal or round telescoping "hatbox" style

Two realistic paths:

Path 1 — Board-game print-on-demand (lowest quantity, fastest):

- **The Game Crafter (US)** — true no-minimum, print-one game manufacturing; custom printed components priced per sheet plus a **per-copy handling fee that starts at \$0.89, drops to \$0.79 at 500 copies and \$0.69 at 1,000 copies**. Offers stock boxes and inserts. Best for prototypes and runs of 1-100. Round/octagonal custom shapes are limited; you may need their Concierge/Laboratory custom services.
- **BoardGamesMaker (BGM)** — premium two-piece rigid boxes, "small print runs from 1 piece upwards" through to full offset production; fully customizable panels. Good bridge between POD and mass production.

Path 2 — Low-MOQ custom rigid/telescoping box manufacturers (better unit cost at 100-500+):

- **QinPrinting** — custom telescope (two-piece lid+base) rigid boxes, **MOQ 500**, 10-20 day production, foil/emboss/spot-UV, FSC options.
- **Chinaprinting4u** — two-piece/telescoping rigid boxes, standard **MOQ 500**, flexible for smaller, offers samples.
- **Tycoon Packaging** — explicitly offers **round hat boxes** and telescoping tube boxes, custom dielines, low MOQs (claims 1 to 10,000), free 3D sample.
- **Kolaxo, Boxit, The Custom Pack** — custom game/rigid boxes, MOQ as low as 50-100, free dieline/3D proofs.
- **Taiwan/China direct** (e.g., Kylinmanufactory) — custom board-game telescoping boxes.

Assessment: For a round "hatbox" telescoping box, the true rigid-box makers (Qin, Chinaprinting4u, Tycoon) give the best quality but effectively want ~500. Below that, use The Game Crafter/BGM. Realistic box cost: The Game Crafter stock box roughly \$2.50-\$4 each at low quantity; custom rigid telescoping box roughly \$2-\$5 each at 500 depending on size and finish. Octagonal is a custom dieline (one-time plate/knife charge) but flat cardboard dielines are cheap to tool compared to metal/plastic.

F. Insert / cradle (die-cut cardboard or moulded pulp)

- **Die-cut cardboard / chipboard insert** is the right low-volume choice: The Game Crafter and every rigid-box maker above can cut a simple channel/tray insert. Custom dielines have only a modest one-time knife cost. For 10-500 units, a folded or die-cut cardboard cradle with a cut-out for the board and two piece pockets prevents rattle at pennies per unit.
- **Moulded pulp: future-only.** Sources confirm pulp requires a custom mould (tooling commonly ~\$5,000-\$20,000, though "low-cost prototype tools" now exist, e.g., Pacific Pulp

Molding) and MOQs typically **1,000–5,000+** (Lianpack: production "usually starts at 1,000–2,000 units"; many Chinese suppliers list 3,000–5,000). Some vendors advertise "low-volume" pulp via cheaper single-cavity prototype tooling, but per-unit cost stays high below ~1,000. **Scope pulp for a later, larger run;** use die-cut cardboard now.

G. Rules / QR card

Trivial. The Game Crafter, BGM, and any local digital printer (or Vistaprint/Moo) print custom cards/booklets at any quantity. A single QR card is cheapest; a folded rules leaflet slightly more. Budget cents per unit. Use a QR code linking to hosted rules/video to minimize printing and allow rule updates without reprinting.

H. Taiwan capability (family-assisted sourcing)

Taiwan is well-suited as a fulfillment/assembly/QC hub and a bridge to tooled production:

- **Small-batch injection molding & mold making:** Intertech (taiwanmoldmaker.com — targets crowdfunding/inventors, molds in 35–40 days, 30–50% deposit), SA Chen (taiwanmold.com — low-volume production, ~40 yrs), Sheng Ching (MOQ ~1,000, accepts small custom/short-lead orders), Chingsui, Boomster (flexible batch sizes, also metal stamping + MIM), Foreshot (injection + metal stamping + assembly under one roof), Pan Taiwan (MIM, stamping, assembly — has served Australian clients).
- **Metal stamping** for a future tooled steel board: Boomster, Foreshot, Intertech.
- **Tinplate litho printing** with small-quantity gang runs: Taiyeng (taiyeng.com.tw, explicitly markets small-quantity metal box printing).
- **Value of family contacts:** on-the-ground QC (critical for magnet-strength consistency and print registration), consolidating components from multiple vendors, hand-assembly of small runs, and negotiating/managing the eventual jump to injection-molded pieces + stamped board once volume justifies tooling. Taiwan's clusters are export-savvy and comfortable with English-language technical files.

Recommendations

Stage 1 — Prototype / first 10–50 units (validate design & market):

1. Tripod pieces: MJF PA12 via Craftcloud or JLC3DP (quote both) + one Australian bureau for fast iteration.
2. Board: Ponoko 430 stainless disc (from \$1.22/disc, choose 1.22 mm) → local Melbourne/Sydney UV flatbed sign shop for the rings. Prototype the graphic as a vinyl label first to lock in the design cheaply.
3. Magnets: sample packs from an Australian supplier (Frenergy/AMF) to tune grip.
4. Box + insert: The Game Crafter or BGM (no minimum), stock or simple custom box + die-cut cardboard cradle.
5. Rules: QR card, printed locally.

Stage 2 — Small commercial run / 100–500 units:

1. Pieces: lock in Craftcloud/JLC3DP MJF at ~\$3-\$7/part; confirm bulk quote.
2. Board: bulk laser-cut discs (nest many per sheet — get AU + SendCutSend/Ponoko quotes) + batch UV flatbed printing.
3. Box: switch to a low-MOQ rigid telescoping/hatbox maker (QinPrinting, Chinaprinting4u, or Tycoon for the round hatbox) at MOQ 500, with a custom die-cut insert included.
4. Magnets: bulk order (thousands) from HSMAG/Stanford or K&J.
5. Use Taiwan family contacts to consolidate/QC/assemble.

Stage 3 — Scale / 1,000+ units (only if demand proven):

1. Injection-mold the tripod pieces in Taiwan (Intertech/SA Chen/Sheng Ching) — break-even vs MJF is ~1,000 units for this part size.
2. Consider stamped + litho-printed or screen-printed steel board (Taiwan stamper + tinplate litho gang run) once board volume is in the thousands.
3. Consider a custom moulded-pulp insert (tooling now amortizes).

Thresholds that change the plan:

- **Pieces → injection molding** when annual volume reliably exceeds ~1,000–2,000 pairs AND the design is frozen (tooling ~\$3,000–\$10,000+ in Taiwan).
- **Board → stamping/litho** when board volume exceeds ~3,000–5,000 (litho minimum ~1,500 sheets; tin tooling 5,000–10,000).
- **Insert → moulded pulp** when volume exceeds ~1,000–2,000 and you want the premium/eco unboxing.

Caveats

- **Grade verification is critical:** most laser shops default to non-magnetic 304/316 stainless. You must explicitly specify 430 (ferritic) or magnetic mild steel, or the pieces won't grip. Ponoko is the confirmed 430 source; others require confirmation.
- **MJF per-part prices are partly synthesized estimates**, not live quotes for your exact geometry. The only hard dated multi-service numbers come from the All3DP July 2024 comparison on similar (not identical) parts, some with premium finishing. Always pull a live instant quote with your actual STL before budgeting.
- **The "\$29 Xometry minimum" and specific JLCPCB minimum fees could not be confirmed** — current sources indicate no hard minimum at either. Verify at quote time.
- **Box MOQs vary and broker marketing claims of "1 to 10,000" should be tested** with an actual quote; true rigid-box tooling economics favor ~500+.
- **Magnet strength must be tuned to board gauge** — thicker steel gives stronger grip; test physically before committing to a magnet size.
- **Moulded pulp low-volume claims** from some vendors rely on cheaper prototype tooling but per-unit cost remains high below ~1,000 units; treat pulp as a scale-up option.
- Import duties/GST and freight from US/China to Australia will affect landed cost; Taiwan-based consolidation can reduce per-shipment overhead.